# Mohamed Elsayed

647-451-4724 — Email — LinkedIn — GitHub — Portfolio

## Education

**University of Toronto** Toronto, Ontario
*Bachelor of Applied Science in Electrical Engineering* *Sep 2025 – May 2030*

- **Relevant Coursework:** Data Structures & Algorithms (Python/C), Linear Algebra, Engineering Math & Computation, Calculus I/II

## Experience

**Technical Lead** Aug 2023 – Mar 2026
*Al Mannar NGO* *Luxor, Egypt*

- Designed and developed production database managing beneficiary records, financial disbursements, and sensitive personal data for charity operations.
- Develop and support backend systems and internal tools in **Python** with relational database, ensuring data integrity and reliability.
- Designed backend infrastructure and database workflows for managing large-scale beneficiary and financial records.

**Undergraduate Research Assistant** July 2025 – Aug 2025
*American University in Cairo (AUC)* *New Cairo, Egypt*

- Built microcontroller-based control system to automate precision translation stage for optics experiments.
- Designed and tested motor control and magnetic field circuits; integrated embedded software in **C++**

**Backend Intern** July 2024 – Aug 2024
*eVision* *New Cairo, Egypt*

- Engineered scalable file-parsing system to automate data ingestion and processing.
- Collaborated with frontend, database, and QA teams in cross-functional agile environment.

**Software Developer & Technical Animator** Oct 2021 – July 2023
*Self-Employed* *Remote*

- Contributed to large-scale game systems supporting **1.7B+ visits** and **250K+ CCU**.
- Developed backend gameplay systems, anti-cheat mechanisms, and animation pipelines using Lua, Python, Blender, and Moon Animator.
- Delivered custom animation and VFX systems through independent commissions for individuals and game projects.

## Projects

**Gaussian Entanglement in Noisy Bosonic Channels** — *Python* Feb 2026

- Derived closed-form entanglement survival threshold for two-mode squeezed vacuum states under symmetric Gaussian channels using symplectic-invariant formulation.
- Implemented numerical benchmarks comparing exact log-negativity with determinant-based analytic upper bound.
- Produced reproducible research note with analytic derivations, parameter sweeps, and publicly available figures & data.

**FPGA Timing Closure & Congestion Prediction System** — *Verilog, Python, PyTorch* April 2026

- Developed graph neural network pipeline to predict FPGA timing violations and routing congestion prior to place-and-route compilation.
- Automated Quartus/Vivado synthesis workflows and parsed timing/utilization reports into structured ML training datasets.
- Represented synthesized FPGA netlists as graphs for critical path and worst negative slack (WNS) prediction using PyTorch Geometric.
- Fine-tuned Qwen-3.5-7B for hardware-aware FPGA timing analysis and optimization tasks using synthesized netlist and timing report data.

**Structural Beam Simulation** — *Python, MATLAB* Nov 2025

- Developed structural beam simulation engine supporting arbitrary cross-sections, distributed loads, and multi-condition boundary constraints using numerical integration methods.
- Implemented automated computation of centroid, neutral axis, area moment of inertia, bending stress, and deflection profiles for custom geometries.
- Built MATLAB-based 3D stress and deformation visualizations with animated load-response simulations under dynamic loading conditions.
- Validated simulated stress and deflection outputs against analytical solutions and physical beam testing measurements.

## Technical Skills

**Programming Languages**: Python, C, C++, Verilog, Java, JavaScript, TypeScript, SQL, MATLAB, HTML/CSS, Lua
**Hardware & Electronics**: FPGA, Microcontrollers, Motor Control, Circuit Design, PCB Assembly, Sensors
**Developer Tools**: Git, CUDA, RoCM, WSL, Codex, Claude Code
**Libraries**: PyTorch, NumPy, Pandas, Matplotlib